

Automatic Generation Control (AGC)

8.22. **EXAMPLE.** Because of the size and complexity of present-day power systems, decentralized control schemes for AGC are more attractive than alternative centralized designs. The main reason is an excessive information requirement imposed by a globally optimal AGC. This has been a prime motivation for building AGC schemes of the decentralized type, where control is confined to individual power areas employing the standard tie-line bias control concept and noninteraction principle, which preserve the steady-state area autonomy and economy of operation (Ćalović *et al.* 1978; Šiljak, 1978). In this kind of control scheme, the areas are represented as overlapping subsystems with the tie-lines being the overlapping parts.

Let us consider a two-area model described in (Ćalović *et al.* 1978) by the equation

$$\dot{x} = Ax + B_1 u_1 + B_2 u_2$$

$$y_i = C_i x, i = 1, 2$$

where the system matrices are specified in the file. The state has the following components: $x = [\Delta x_1^T, \Delta v_1, \Delta p_e, \Delta v_2, \Delta x_2]^T$, where Δx_i is the deviation of the internal states of area i with respect to their nominal value, Δv_i are variables introduced to achieve integral action, and Δp_e is the variation of the total power exchange between the areas with respect to its nominal value.

Problem:

1. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time of choice, compatible with the continuous-time system dynamics). Perform the following analysis:
 - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
 - b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
2. For different control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
 - a. Compute the continuous-time fixed modes
 - b. Compute the discrete-time fixed modes

- c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
- d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
- e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectory (generated both in continuous-time and in discrete-time) of the variable Δp_e starting from a common random initial condition.